



On the stretch factor of Delaunay triangulations of points in convex position [☆]

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ABSTRACT

Let S be a finite set of points in the Euclidean plane. Let G be a geometric graph in the plane whose point set is S . The *stretch factor* of G is the maximum ratio, among all points p and q in S , of the length of the shortest path from p to q in G over the Euclidean distance $|pq|$. Keil and Gutwin in 1989 [11] proved that the stretch factor of the Delaunay triangulation of a set of points S in the plane is at most $2\pi/(3 \cos(\pi/6)) \approx 2.42$. Improving on this upper bound remains an intriguing open problem in computational geometry. In this paper we consider the special case when the points in S are in convex position. We prove that in this case the stretch factor of the Delaunay triangulation of S is at most $\rho = 2.33$.

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1. Introduction

Let S be a finite set of points in the Euclidean plane. A *Delaunay triangulation* of S is a triangulation in which the circumscribed circle of every triangle contains no point of S in its interior [6]. It is well known that if the points in S are in *general position* (i.e., no four points in S are cocircular) then the Delaunay triangulation of S is unique [6]. To simplify the discussion, we shall assume that the Delaunay triangulation is unique, even though the results in this paper are not contingent on this assumption.

An alternative equivalent definition of *Delaunay triangulation* is:

Definition 1. (See [6].) An edge xy is in the Delaunay triangulation of S if and only if there exists a circle through points x and y whose interior is devoid of points of S .

Chew [5] showed a lower bound of $\pi/2$ on the stretch factor of Delaunay triangulations.⁴ This lower bound of $\pi/2$ [5] was widely believed to be tight (for example, see page 470 in [14]) until very recently (2009) when Bose et al. [3] showed that the stretch factor of the Delaunay triangulation of a point set is at least 1.5846. They also showed a slightly worse lower bound of 1.581 for point sets in convex position. Dobkin, Friedman, and Supowit [7,8] in 1987 showed that the Delaunay

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⁴ In the same paper [5], Chew proved that the stretch factor of a Delaunay triangulation under the L_1 norm is at most $\sqrt{10}$.

triangulation of a point-set S has stretch factor $(1 + \sqrt{5})\pi/2 \approx 5.08$. This ratio was improved by Keil and Gutwin [11,12] in 1989 to $2\pi/(3\cos(\pi/6)) \approx 2.42$, which currently stands as the best upper bound on the stretch factor of Delaunay triangulations. In an open-problem session of the 19th Canadian Conference on Computational Geometry (CCCG 2007) [2], Bose suggested looking at the special case when the points in S are in convex position.

In addition to its theoretical interest, improving the current upper bound on the stretch factor Delaunay triangulations has a direct impact on the problem of constructing geometric *spanners* of Euclidean graphs, which has significant applications in the area of wireless computing (for more details, see the following book on spanners [14]). Many spanner constructions in the literature rely on extracting subgraphs of the Delaunay triangulation (see for example [4,9,10,13]), and hence improving the upper bound on the stretch factor of Delaunay triangulations will automatically improve the upper bound on stretch factors of all such spanners.

In this paper we show that the stretch factor of the Delaunay triangulation of a point-set in convex position is at most $\rho = 2.33$. This upper bound is smaller than the current best upper bound on the stretch factor of the Delaunay triangulation of a point-set in general position, which is approximately 2.42.

2. The results

Let $\rho = 2.33$. This section is devoted to proving the following theorem:

Theorem 1. *Let S be a finite set of points in convex position in the plane. The stretch factor of the Delaunay triangulation of S is at most ρ .*

Proof. Let D be the Delaunay triangulation of S . Let $p, q \in S$ be two arbitrary points, and denote by $|pq|$ the Euclidean distance between p and q . Denote by $P(p, q)$ a shortest path between p and q in D , and by $|P(p, q)|$ the *weight* of $P(p, q)$, that is, the sum of the Euclidean distances between every two consecutive points on $P(p, q)$. We shall prove that $|P(p, q)| \leq \rho|pq|$. Without loss of generality, we shall assume that the line pq is horizontal.

We proceed by induction on the rank of $|pq|$ among all pairs of points in S . (We assume that ties are broken arbitrarily.) If the distance between p and q is the smallest among all pairs of points in S , then the circle with diameter pq contains no points of S in its interior. By Definition 1, pq is an edge in D , and hence $|P(p, q)| = |pq| \leq \rho|pq|$.

Now suppose that the statement is true for any pair of points whose distance is less than $|pq|$. Assume that pq is not an edge in D (otherwise we are done by the same argument as above).

Since the point-set S is convex, there exists a supporting line L_p through p such that all other points of S lie on one side of L_p [15]. Let L_q be a line through q that is parallel to L_p . Then it is easy to see that on one side of the line pq —either above or below it—all points of S lie between L_p and L_q . Without loss of generality, assume that all points of S above the line pq lie between L_p and L_q . Note that if this set of points is empty, then p and q are connected in D by a horizontal path (possibly a single edge) of weight $|pq|$, and the statement follows.

Let $t \in S$ be a point above pq that maximizes the angle $\gamma = \angle ptq$. We need the following lemma:

Lemma 2. (See [12], Lemma 1.) *Let C be a circle passing through two points p and q . If no point of S lies inside C above pq , then there is a path from p to q in D whose length is at most the length of the lower arc of C between p and q .*

Let C be the circle passing through p, t, q . Since the angle $\gamma = \angle ptq$ is maximized, no point of S lies inside C above pq . By Lemma 2, there exists a path from p to q in D whose length is at most $2r\gamma$, where r is the radius of C . Since $|pq| = 2r \sin \gamma$, this implies that if $\gamma \leq 2.058$, then there is a path from p to q in D whose length is at most $2r\gamma = (\gamma/\sin \gamma)|pq| \leq 2.33|pq| = \rho|pq|$, and we are done.⁵

Therefore, we can assume that $\gamma > 2.058 > \pi/2$ in the rest of the proof. Consequently, $|pt| < |pq|$ and $|tq| < |pq|$, and by the inductive hypothesis, we have

$$|P(p, t)| \leq \rho|pt| \tag{1}$$

and

$$|P(t, q)| \leq \rho|tq|. \tag{2}$$

The following lemma shows that if p and t or t and q are connected by a direct edge in D , then we are done.

Lemma 3. *If $|P(p, t)| = |pt|$ or $|P(t, q)| = |tq|$, then $|P(p, q)| \leq \rho|pq|$.*

⁵ All angles are measured in radians in this paper.

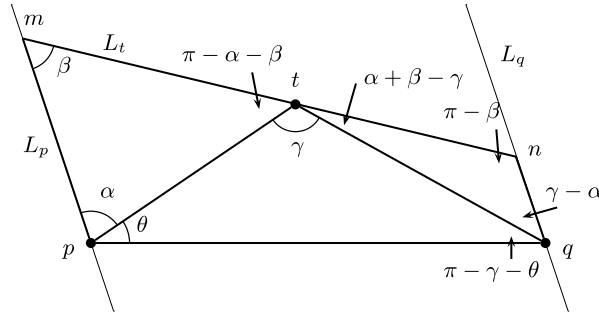


Fig. 1. An illustration of the structure above pq .

Proof. Suppose that $|P(p, t)| = |pt|$, and let $\theta = \angle tpq$. In the triangle $\triangle ptq$ we have $\frac{|pt|}{|pq|} = \frac{\sin(\pi - \gamma - \theta)}{\sin \gamma} = \frac{\sin(\gamma + \theta)}{\sin \gamma}$, and $\frac{|tq|}{|pq|} = \frac{\sin \theta}{\sin \gamma}$. Hence

$$\frac{|P(p, q)|}{|pq|} \leq \frac{|P(p, t)| + |P(t, q)|}{|pq|} \leq \frac{|pt| + \rho|tq|}{|pq|} = \frac{\sin(\gamma + \theta) + \rho \sin \theta}{\sin \gamma} = \cos \theta + \frac{\sin \theta (\cos \gamma + \rho)}{\sin \gamma}. \quad (3)$$

The last equality follows from standard trigonometric identities. Define the function $g(\gamma)$ of γ in the interval $(2.058, \pi)$ as follows: $g(\gamma) = (\cos \gamma + \rho) / \sin \gamma$. Since $\rho = 2.33$ and $2.058 < \gamma < \pi$, we have $g'(\gamma) = (-1 - \rho \cos \gamma) / \sin^2 \gamma > (-1 - \rho \cos 2.058) / \sin^2 \gamma > 0$, and hence $g(\gamma)$ is an increasing function in the chosen interval. Since $\gamma < \pi - \theta$, we have $g(\gamma) < g(\pi - \theta)$. Therefore, $g(\gamma) = (\cos \gamma + \rho) / \sin \gamma < (\cos(\pi - \theta) + \rho) / \sin(\pi - \theta) = (-\cos \theta + \rho) / \sin \theta$. The last inequality, together with inequality (3), gives $|P(p, q)| / |pq| \leq \cos \theta + g(\gamma) \sin \theta \leq \rho$.

The proof is analogous when $|P(t, q)| = |tq|$. $\square$

We may assume that t does not lie on L_p because otherwise, tp is in an edge of the convex hull of S and hence is in D , and by Lemma 3, we are done. Similarly, we may assume that t does not lie on L_q .

Since the point-set S is convex, there exists a supporting line L_t passing through t such that all other points in S are below L_t . Similar to the above, we can assume that the line L_t does not go through p and q because otherwise by Lemma 3, we are done.

Let m and n be the intersections of L_t with L_p and L_q , respectively. By the above discussion, m and n are above the line pq and the non-degenerate triangles $\triangle pmt$, $\triangle tpq$, and $\triangle tqn$ do not overlap. Let $\alpha = \angle mpt$, and $\beta = \angle pmt$. See Fig. 1 for illustration.

By the above discussion, t is above pq , and hence

$$0 < \theta < \pi - \gamma \quad \text{and} \quad \gamma < \pi.$$

Since t does not lie on L_p , we have $\alpha > 0$, $\beta > 0$. Since t does not lie on L_q , we have $\gamma - \alpha > 0$. Because m and p do not overlap, we have $\alpha + \beta < \pi$. Putting all together, and recalling that $2.058 < \gamma$, we have

$$0 < \alpha < \gamma < \pi,$$

$$0 < \beta < \pi - \alpha,$$

and

$$2.058 < \gamma < \pi.$$

Since $pt \notin D$, and by convexity of S , there exists a path in D from p to t that is convex-away from pt and that lies in the triangle $\triangle pmt$. By convexity, the length of this path is at most $|pm| + |mt|$ (see [1, p. 42]). Therefore, the length of the shortest path from p to t is at most $|pm| + |mt|$. Similarly, the length of the shortest path from t to q is at most $|tn| + |nq|$. Let

$$\rho_1 = \frac{|pm| + |mt|}{|pt|}$$

and

$$\rho_2 = \frac{|tn| + |nq|}{|tq|}.$$

Then

$$|P(p, t)| \leq \rho_1 |pt| \quad (4)$$

and

$$|P(t, q)| \leq \rho_2 |tq|. \quad (5)$$

Combining inequalities (1), (2), (4), and (5), we have:

$$\frac{|P(p, q)|}{|pq|} \leq \frac{|P(p, t)| + |P(t, q)|}{|pq|} \leq \frac{\min(\rho_1, \rho) |pt| + \min(\rho_2, \rho) |tq|}{|pq|},$$

and since $\frac{|pt|}{|pq|} = \frac{\sin(\gamma + \theta)}{\sin \gamma}$ and $\frac{|tq|}{|pq|} = \frac{\sin \theta}{\sin \gamma}$ in $\triangle ptq$, we have

$$\frac{|P(p, q)|}{|pq|} \leq \frac{\min(\rho_1, \rho) \sin(\gamma + \theta) + \min(\rho_2, \rho) \sin \theta}{\sin \gamma}. \quad (6)$$

Now we have

$$\rho_1 = \frac{|pm| + |mt|}{|pt|} = \frac{\sin(\pi - \alpha - \beta) + \sin \alpha}{\sin \beta} = \sin \alpha \left(\frac{1 + \cos \beta}{\sin \beta} \right) + \cos \alpha \quad (7)$$

and

$$\rho_2 = \frac{|tn| + |nq|}{|tq|} = \frac{\sin(\gamma - \alpha) + \sin(\alpha + \beta - \gamma)}{\sin(\pi - \beta)} = \sin(\gamma - \alpha) \left(\frac{\sin \beta}{1 + \cos \beta} \right) + \cos(\gamma - \alpha). \quad (8)$$

Now combining (7) with (8) we get

$$\rho_2 = \sin(\gamma - \alpha) \left(\frac{\sin \alpha}{\rho_1 - \cos \alpha} \right) + \cos(\gamma - \alpha). \quad (9)$$

We proceed by distinguishing two cases: (i) $\rho_1 \geq \rho$ or $\rho_2 \geq \rho$, and (ii) $1 \leq \rho_1, \rho_2 \leq \rho$. The following two lemmas show that $\frac{|P(p, q)|}{|pq|} \leq \rho$ in both cases. That is all we need since $\rho_1, \rho_2 \geq 1$.

Lemma 4. If $\rho_1 \geq \rho$ or $\rho_2 \geq \rho$, then $\frac{|P(p, q)|}{|pq|} \leq \rho$.

Proof. Suppose that $\rho_1 \geq \rho$. By equality (9) we have:

$$\begin{aligned} \rho_2 &= \sin(\gamma - \alpha) \left(\frac{\sin \alpha}{\rho_1 - \cos \alpha} \right) + \cos(\gamma - \alpha) \\ &\leq \sin(\gamma - \alpha) \left(\frac{\sin \alpha}{\rho - \cos \alpha} \right) + \cos(\gamma - \alpha) \end{aligned} \quad (10)$$

$$\begin{aligned} &= \frac{\sin(\gamma - \alpha) \sin \alpha + \rho \cos(\gamma - \alpha) - \cos(\gamma - \alpha) \cos \alpha}{\rho - \cos \alpha} \\ &= \frac{\rho \cos \gamma \cos \alpha + \rho \sin \gamma \sin \alpha - \cos \gamma}{\rho - \cos \alpha} \\ &= -\rho \cos \gamma + \frac{(\rho^2 - 1) \cos \gamma + \rho \sin \gamma \sin \alpha}{\rho - \cos \alpha} \\ &\leq -\rho \cos \gamma + \frac{(\rho^2 - 1) \cos \gamma + \rho \sin \gamma}{\rho - \cos \alpha}. \end{aligned} \quad (11)$$

Inequality (10) is true because $\sin \alpha > 0$, $\sin(\gamma - \alpha) > 0$, $\rho - \cos \alpha > 0$, and $\rho_1 \geq \rho$. Inequality (11) is true because $\sin \gamma > 0$, $\rho - \cos \alpha > 0$, and $\sin \alpha \leq 1$.

Define the function f of γ as follows: $f(\gamma) = (\rho^2 - 1) \cos \gamma + \rho \sin \gamma$. Since $\rho = 2.33$ and $2.058 < \gamma < \pi$, we have $f'(\gamma) = -(\rho^2 - 1) \sin \gamma + \rho \cos \gamma < 0$ and thus $f(\gamma)$ is a decreasing function of γ in the interval $(2.058, \pi)$. Therefore $f(\gamma) = (\rho^2 - 1) \cos \gamma + \rho \sin \gamma \leq (2.33^2 - 1) \cos(2.058) + 2.33 \sin(2.058) \leq 0$. Also note that $\rho - \cos \alpha > 2.33 - 1 > 0$, and hence

$$\rho_2 \leq -\rho \cos \gamma + \frac{\cos \gamma (\rho^2 - 1) + \rho \sin \gamma}{\rho - \cos \alpha} \leq -\rho \cos \gamma. \quad (12)$$

Applying inequality (12) to inequality (6), we obtain:

$$\begin{aligned}
 \frac{|P(p, q)|}{|pq|} &\leq \frac{\min(\rho_1, \rho) \sin(\gamma + \theta) + \min(\rho_2, \rho) \sin \theta}{\sin \gamma} \\
 &\leq \frac{\rho \sin(\gamma + \theta) + \rho_2 \sin \theta}{\sin \gamma} \\
 &\leq \frac{\rho \sin(\gamma + \theta) - \rho \cos \gamma \sin \theta}{\sin \gamma} \\
 &= \frac{\rho \sin \gamma \cos \theta + \rho \cos \gamma \sin \theta - \rho \cos \gamma \sin \theta}{\sin \gamma} \\
 &= \rho \cos \theta \\
 &\leq \rho.
 \end{aligned} \tag{13}$$

By symmetry, the same holds true if $\rho_2 \geq \rho$. This completes the proof of Lemma 4. $\square$

Lemma 5. If $1 \leq \rho_1, \rho_2 \leq \rho$, then $\frac{|P(p, q)|}{|pq|} \leq \rho$.

Proof. Since $\rho_1, \rho_2 \leq \rho$, by inequality (6) we have:

$$\frac{|P(p, q)|}{|pq|} \leq \frac{\min(\rho_1, \rho) \sin(\gamma + \theta) + \min(\rho_2, \rho) \sin \theta}{\sin \gamma} = \frac{\rho_1 \sin(\gamma + \theta) + \rho_2 \sin \theta}{\sin \gamma}.$$

By equality (9), we have:

$$\frac{|P(p, q)|}{|pq|} \leq \frac{\rho_1 \sin(\gamma + \theta) + \frac{\sin(\gamma - \alpha) \sin \alpha \sin \theta}{\rho_1 - \cos \alpha} + \cos(\gamma - \alpha) \sin \theta}{\sin \gamma}.$$

For any fixed values of α, γ, θ , define the function $h(\rho_1)$ of ρ_1 in the interval $[1, \rho]$ as follows:

$$\begin{aligned}
 h(\rho_1) &= \rho_1 \sin(\gamma + \theta) + \frac{\sin(\gamma - \alpha) \sin \alpha \sin \theta}{\rho_1 - \cos \alpha} \\
 &= \sin(\gamma + \theta) \left(\rho_1 - \cos \alpha + \frac{\sin(\gamma - \alpha) \sin \alpha \sin \theta}{\sin(\gamma + \theta)(\rho_1 - \cos \alpha)} \right) + \sin(\gamma + \theta) \cos \alpha.
 \end{aligned}$$

Let $C_1 = \sin(\gamma + \theta)$, $C_2 = \frac{\sin(\gamma - \alpha) \sin \alpha \sin \theta}{\sin(\gamma + \theta)}$, and $C_3 = \sin(\gamma + \theta) \cos \alpha$. Then

$$h(\rho_1) = C_1 \left(\rho_1 - \cos \alpha + \frac{C_2}{\rho_1 - \cos \alpha} \right) + C_3$$

and its derivative is

$$h'(\rho_1) = C_1 \left(1 - \frac{C_2}{(\rho_1 - \cos \alpha)^2} \right).$$

Since $C_1, C_2, C_3 > 0$ are fixed and $\rho_1 - \cos \alpha > 0$, $h'(\rho_1)$ is a monotonically increasing function in the interval $[1, \rho]$. Thus by the *first derivative test* in calculus, the maximum value of the function $h(\rho_1)$ in the interval $[1, \rho]$ occurs on the boundary. Hence the maximum value of $\frac{|P(p, q)|}{|pq|}$ also occurs on the boundary of the interval $[1, \rho]$. When $\rho_1 = 1$, we have $|P(p, t)| = |pt|$ and by Lemma 3 we have $\frac{|P(p, q)|}{|pq|} \leq \rho$; and when $\rho_1 = \rho$, by Lemma 4 we have $\frac{|P(p, q)|}{|pq|} \leq \rho$. In either case, we have $\frac{|P(p, q)|}{|pq|} \leq \rho$. This completes the proof of Lemma 5. $\square$

Combining Lemmas 4 and 5, we conclude the proof of Theorem 1. $\square$

3. Concluding remarks

In this paper we showed that the stretch factor of the Delaunay triangulation of a point-set in convex position is at most 2.33. Our upper bound results, together with the recent lower bound results by Bose et al. [3], demonstrate that the special case when the point-set is in convex position is worth further investigation. While studying this simpler case may not lead to improved upper bounds for the general case, it may shed some light on the intrinsic difficulty of this problem.

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